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Video Traffic Analysis over LEACH-GA routing protocol in a WSN

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Abstract

One of the most advanced evolutions of Wireless communication is Wireless Sensor Network (WSN), which plays a vital role in many applications. The growth and attention towards the designing of many new QoS aware routing protocol for WSN has been increased for the past few decades. Energy awareness has a greater significance while developing a routing protocol in order to increase network lifetime. In this paper, a traditional LEACH-GA routing protocol has been used for analyzing the video traffic over the sensor network.

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Keywords: Video transmission; Hierarchical protocol; LEACH-GA; Routing protocol; Energy Efficient; Genetic Algorithm; H.264/AVC;

1. Introduction

Wireless Sensor Network (WSN) is one of the most important technologies that plays a major role on Surveillance [2], remote monitoring [3] and controlling [1], target tracking, environmental monitoring, disaster relieves, agriculture [4] and so on. Due to the rapid increasing demand [7] on Wireless Multimedia Network (WMSN), it is

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important to provide Quality of Service (QoS) aware routing protocol for the enhancement of energy efficient and network stability.

2. Related Work

In WSN, the major challenging and on-going research issue is network lifetime and energy consumption. Low Energy Adaptive Cluster hierarchical (LEACH) [10] was the first hierarchical routing protocol especially designed for random distribution of nodes with setup and steady state phases in order to reduce the energy consumption of nodes. However, LEACH selects the cluster heads (CHs) randomly which reflects and degrades the network performance. In Centralized LEACH (LEACH-C) [11] with the prior knowledge of location and node energy, selections of CHs are improved when compared to LEACH protocol. Another routing protocol which is used minimum spanning tree technique for the selection of CHs is LEACH-E [12]. Pre stage is defined before setup and steady state phase in IBLEACH [13] to enhance the performance of network by reducing the energy consumption. Video streaming through zigbee technologies based wireless sensor network was proposed by [14]. Acceptable video quality has been achieved by transmitting a video over zigbee network. Quality of Service (QoS) can be analysed in WSN by transmitting video packets over the network [15][6].

3. Problem Statement and Our Contribution

In general, based on the network structure the routing protocols can be broadly classified into three main categories, namely, flat routing, location-based routing, and hierarchical based routing protocol. Among these hierarchical routing protocols has more energy-efficient and reliable when compared with other protocols [16]. One of the traditional Genetic Algorithm based hierarchical protocols is LEACH-GA. However, LEACH-GA supports for many WSN applications for data analysis. Our main goal to prove that LEACH-GA, will not only supports for data analyze but also we may use these routing protocols for video traffic analyzes over WSN. Our result shows that traditional LEACH-GA routing protocol supports video transmission over WSN with an acceptable video quality of PSNR and Jitter values.

4. LEACH-GA Routing Protocol

In LEACH-GA [4], a new phase is presented before the setup and steady state phase namely preparation phase which help to select the optimal probability of cluster head using genetic algorithm. This preparation phase is carried out only once at the beginning by message sending from base station (BS), in which genetic algorithm helps to determine the optimal probability to become a cluster head (CH) for completing the initial round. The optimal cluster probability selection will also depend on nodal energy of the network. After the completion of Preparation phase, setup phase begins by transmitting a message from BS to CHs. However, threshold value T (n) is used for CH selection which depends on cluster head optimal probability p, current rounds of the network r, number of non-cluster heads in 1/p rounds.

$$T(n) = \begin{cases} \frac{p}{1-p(r \bmod \frac{1}{p})} & \forall n \in G \\ 0 & if\ n \notin G \end{cases} \tag{1}$$

Network gives a random number between 0's and 1's for all nodes. Cluster Heads are determined if the generated number is less than T (n). Apart from CHs, the rest of the nodes will acts as a non-CHs or cluster members (CM). Then steady state phase begins to form a cluster for data transmission by sending a request message to all CMs from the CHs. After the successful formation of cluster, the data's are transmitted from source to destination through CMs and CHs. First radio model was used to analyse the LEACH-GA protocols. Cluster head energy consumption per node is,

$$E_{CH\,(l,d)} = \begin{cases} l \times [E_{elec}\left(\frac{n}{k}-1\right) + E_{DA}\frac{n}{k} + E_{elec} + \varepsilon_{fs} \times d_{to\,BS}^{2} & if\ d_{to\,BS} < d_0 \\ l \times [E_{elec}\left(\frac{n}{k}-1\right) + E_{DA}\frac{n}{k} + E_{elec} + \varepsilon_{mp} \times d_{to\,BS}^{4} & if\ d_{to\,BS} \geq d_0 \end{cases} \tag{2}$$

$$d_0 = \sqrt{\frac{\varepsilon_{fs}}{\varepsilon_{mp}}} \tag{3}$$

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2} \tag{4}$$

Where ‘d_0’ represents the threshold distance, ‘n’ denotes the number of sensor nodes, ‘k’ indicates the number of clusters, ‘l’ is the length of message and ‘d’ is the distance between nodes to BS.
Similarly, non-cluster head energy consumption by nodes is,

$$E_{non-CH}(l.d) = l \times E_{elec} + l \times \varepsilon_{fs} \times d^2_{to\,CH} \tag{5}$$

Overview of LEACH-GA Algorithm:
Begin
Step 1: Initialize
Network Parameters
Probability
Energy Parameters
Population
GA starts
Step 2: Compute fitness function for CHs selection
Step 3: Use Roulette Wheel method for best individual selection
Step 4: Apply Crossover and Mutation operation
Step 5: Compare evaluated fitness value with the initial value
Step 6: If
Fitness value < Initial value
Update Initial Population by repeating From Step 2
else
Go to step 7
end
GA end
Step 7: Data transfer between nodes, through BS and CHs using relay nodes
Stop

5. H.264/AVC Video Codec

H.264/AVC is a new and well established video coding standard recommended by ITU-T or International Standard 14496-10 or MPEG -4 part 10 Advanced Video Coding (AVC) of ISO/IEC [5][6].

5.1. Video Encoder

An H.264/AVC produces a compressed bit streams for the input video with prediction, DCT transform and encode process as shown in Figure 1.

5.2. Video Decoder

In order to reproduce video sequence, H.264 decoder accomplishes decoding, Inverse DCT transform and reconstruction to recover the input video

6. Video transmission over LEACH-GA

Video transmission through hierarchical routing protocol is one of the major challenging tasks in wireless sensor network. LEACH-GA based video transmission in WSN through H.264/AVC, which includes the following steps.

1. Creating a WSN with ‘N’ number of nodes.
2. Assume ‘M’ number of nodes as video sensor nodes among the ‘N’ – sensor nodes and also assume Base station (BS) at the centre of the network.

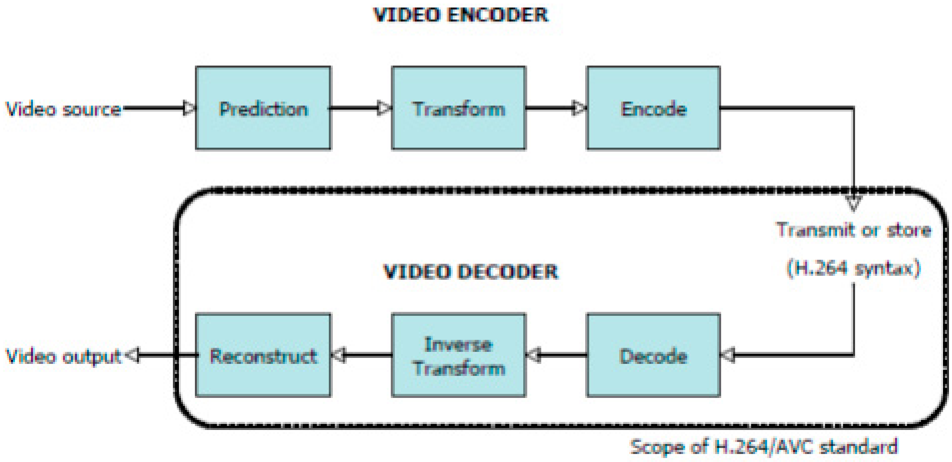

Figure 1: H.264/AVC Video Codec [9]

3. Perform LEACH-GA to find the shortest routing between the source and destination video sensor node through Cluster Heads (CHs) and BS.
4. After finding the shortest route, video are transmitted based on H.264/AVC encoder technique through the network.
5. At the receiver node, H.264/AVC decoder technique is used to reconstruct the transmitted video.
6. Finally, from the received video, jitter value, average peak signal to noise ratio (PSNR), total time taken are computed to analyse the received video quality through the sensor network.

7. Simulation Result

The analyzing of video transmission over LEACH-GA [4] protocol was simulated by using standard MATLAB tool. Randomly generated nodes are spread over an area of 100 x 100 m^2 with center BS as shown in Figure 3. The cluster heads are represented as cross mark and rectangular red box represents the video sensor nodes.Figure 3 shows the network structure in which the video has been transmitted over a WSN with a standard H.264 encoding and decoding techniques. Pink dark line represents the transmission path between source and destination. Among the 10 video sensor nodes 13 and 82 are selected as a source and destination nodes respectively.

Our previous work [8], conclude that LEACH GA enhance the network lifetime by 54% and 47% when compared to LEACH and LEACH C protocols as shown in Figure 2. Let us consider 2 road surveillance videos for the simulation, namely video 1 (Figure 4) and video 2 (Figure 5) are used to transmit over the network. Table 1 and Table 2 represent the video and simulation parameters of the network

Table 1: Video Parameters

Parameters	Video 1	Video 2
Bits Per Pixels	24	24
Frame Rate	30	10
Height of the Frame	176	240
Width of the Frame	320	320

Number of Frames	15	15
Video Format	RGB24	RGB24

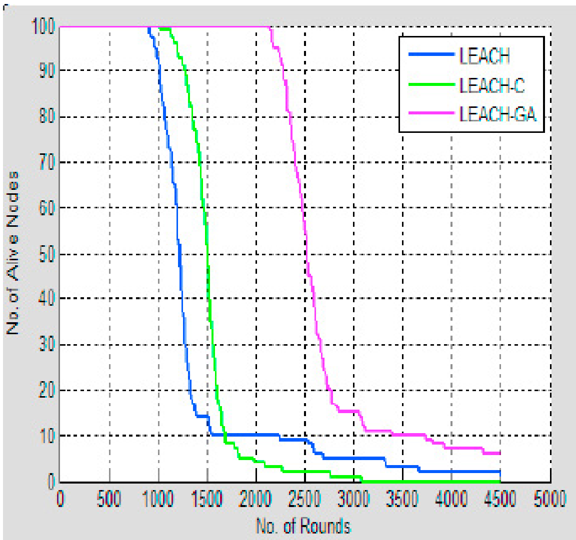

Figure 2: Comparison of LEACH-GA protocol with LEACH and LEACH C [8]

Table 2: Simulation Parameters

Network Parameters	Values
Network Size	100 x 100 m^2
Total number of nodes	100
Number of Video nodes	10
Video sensor Nodes	82 91 13 92 64 10 28 55 96 97
Routing Protocols	LEACH-GA
Initial Energy, E_o	0.5J/nodes
Energy for transmitting the data, E_{TX}	50nJ/bit
Energy for receiving the data, E_{RX}	50nJ/bit
Short distance Amplification Energy, E_{fs}	10pJ/bit/m^2
Long distance Amplification Energy, E_{mp}	0.0013pJ/bit/m^2
Data Aggregation Energy, E_{da}	5nJ/bit
Probability of Cluster Head, p	0.1, 0.15 and 0.5
Total number of Iteration	10

The performances of the received video are analysed with the parameters of PSNR, and Jitter. Squared mean error value between the video transmitted and reconstructed/received video can be calculated by Peak Signal to Noise Ratio (PSNR) and Mean Square Error (MSE). Usually the PSNR is measured in dB and also the higher value of PSNR (>20dB) results in good quality image. For video files, PSNR is determined by adding up the squared error value for each and every individual frames. Our simulation results shows that received Average PSNR value as 34.9 dB and 33.7 dB for video 1 and video 2 respectively, which results in good quality of video file transmission through LEACH-GA routing protocol. Figure 6 shows the PSNR values for individual frames that are transmitted through the network. Time delay between the arrivals of received video packets can be measured by a parameter called Jitter. Normally the acceptable jitter value for video transmission is not less than 10ms and greater than 50ms (≤ 10ms – 50 ms). For the video 1 and video 2, the value of jitter is 18 milliseconds and 43 milliseconds, negative sign in the simulation results shows that the values are measured in milli seconds (ms). The average time taken for transmitting the video1 and video 2 are 133 sec and 210 sec respectively as shown in Figure 7.

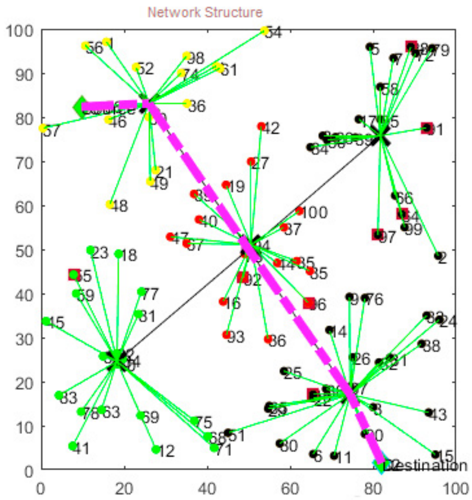

Figure 3: Network Structure

Figure 4: Video 1

Figure 5: Video 2

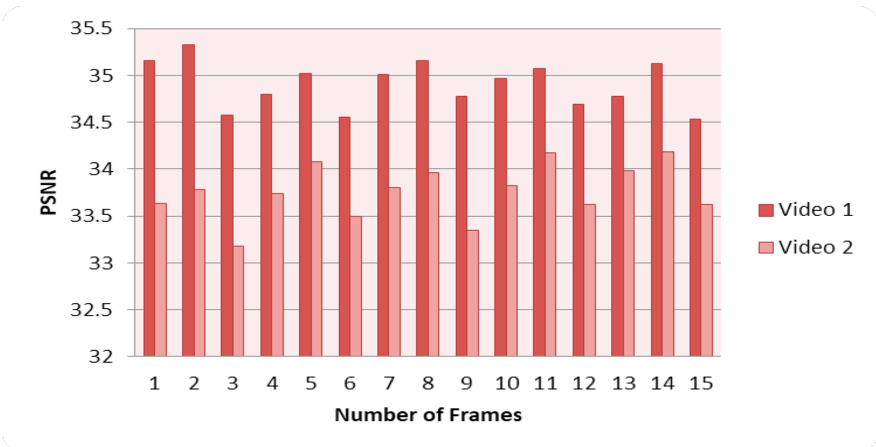

Figure 6: Comparison of PSNR values

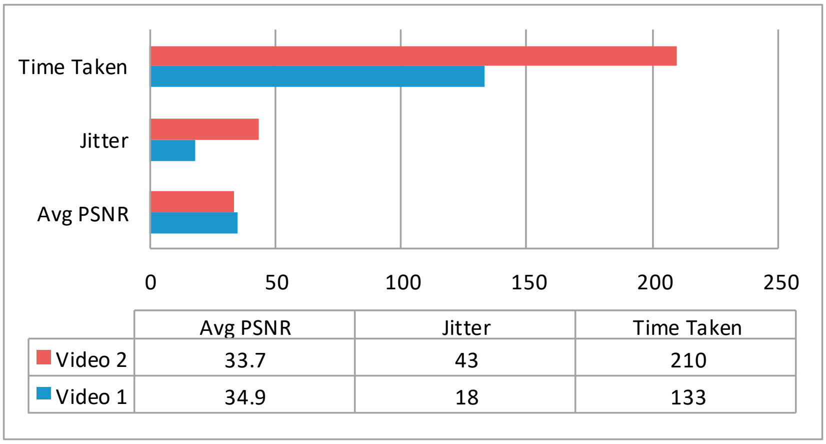

Figure 7: Comparison of video 1 and video 2 simulation parameters

8. Conclusion

A video traffic analysis over the WSN network with the acceptable video quality parameters like PSNR, and jitter has been achieved by using a traditional LEACH-GA routing protocol. Further, the work can be extended towards the real time video transmission using LEACH-GA routing protocol for analyzing the video parameters to improve the network performance.

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